ODE Exam One Review on the Plane

- Autonomous differential equations, phase portraits
- Solving First Order Differential Equations (separable, method of integrating factor)
- 2d Systems of the form $\vec{x}'(t) = A\vec{x}(t)$

Autonomous differential equations

- Recall a differntial equation is some equation that involves a function f usually mapping $\mathbb{R} \to \mathbb{R}$ as well as its derivative
  - In this class usually we denote by $y(t)$ such a function $\mathbb{R} \to \mathbb{R}$
  - – PDEs consider more general maps and can impose relations between partials
- We say a differential equation is **autonomous** if it's time invariant, in other words if we have

$$f'(t) = g(f(t))$$

for some function g. This notably means that the rate of change is entirely determined by the current function value and does not depend on the input/time

- This means that if we draw it like a vector field the horizontal strips all have the same direction, we can only draw it like this because we can put y on the x axis which we can't do in general
- for example $y' = y(1-y)$ is an autonomous differential equation, but $y'(t) = ty(t)$ isn't
- Notice that for any c such that $g(c) = 0$ we have that the constant function $y = c$ is a solution to this differential equation
  - Because of time invariance, the behavior of g (in particular its sign) between these constant solutions lets us conclude any other solution must converge to one of these guys because we know $f'(t)$

Solving First Order ODEs

- A first order ODE is an equation of the form

$$y' = f(y, t)$$

- where the order denotes the highest derivative involved. This is more general than we'll actually be able to solve though
- If f can be factored into a function of y and a function of t then we say the ODE is separable, in this case

$$y' = p(y)q(t) \to \frac{1}{p(y)}y' = q(t)$$

- and we can integrate using the chain rule (but remember to check for lost solutions)
- We say a differential equation is **linear** if the terms involving derivatives only use linear functions in $y^{(k)}$ (so they're allowed to depend on t)
  - For example $y'' = \sin(y)$ is not allowed but $y'' = \sin(t)y'$ is allowed.
  - Notably separable differential equations need not be linear unless p is the identity (we can't always write some function $p(t)y + q(y)$ as the product of functions in t, y
- We mainly focus on first order linear differential equations, we say the standard form for such an ODE is

$$y' + p(t)y = q(t)$$

- We solve these equations via the method of the integrating factor $\mu = \exp(\int p(t)dt))$, observe that multiplying this through gives that

$$\mu y' + \mu p(t)y = \mu q(t)$$

- where $\mu' = \mu p(t)$ so by the product rule we can write this as

$$(\mu y)' = \mu q(t) \to y = \frac{1}{\mu} \int \mu q(t) dt$$

Two Dimensional ODEs

- Okay we generalize the above to systems that involve two functions $x_1(t), x_2(t)$ and write $\vec{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$
- We study two dimensional ODEs with constant coefficients, so systems of the form

$$x_1{}' = ax_1 + bx_2$$

$$x_2{}' = cx_1 + bx_2$$

so we can write this as $\vec{x}'(t) = Ax(t)$

- There's some way to write a second order ODE (with constant coefficients) as a two dimension first order ODE, I think you take $x_1(t) = y$ and $x_2(t) = y'$ or something like this
- Not sure where these equations come from but solve for the eigenvalues and eigenvectors; our solutions are of the form $ce^{\lambda t}v$ for each pair
  - If there's just one then we form the second solution with $ce^{\lambda t}(tv + w)$ I guess just remember this as the intuitive thing being to try tv but for some reason we need to add w for this to actually work
  - w is such that $(A - \lambda I)w = v$, so $A - \lambda I$ kills v but can only kill w after two applications
- For the real eigenvalue case you just draw out our two solutions and draw the arrows according to the sign of λ
  - Then the intermediate solutions get drawn towards the fast one more or less, but I think I will need to review all the pictures at some point